

Exercise of Supervised Learning: Boosting Part 1

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Exercise 1: AdaBoost - Empirical Risk

Let $\hat{f}(\mathbf{x}) = \sum_{m=1}^M \beta^{[m]} \hat{b}^{[m]}(\mathbf{x})$ be the scoring function after running AdaBoost for $M \in \mathbb{N}$ iterations. Show that the average empirical risk (on $\mathcal{D}_{\text{train}}$) of the corresponding classifier $h(\mathbf{x}) = \text{sign}(\hat{f}(\mathbf{x}))$ is bounded as follows

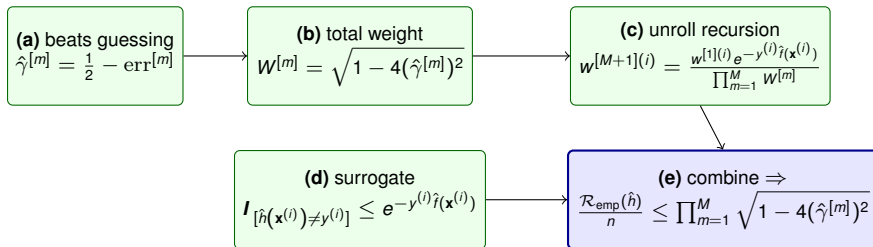
$$\frac{\mathcal{R}_{\text{emp}}(\hat{h})}{n} = \frac{\sum_{i=1}^n \mathbf{1}_{[\hat{h}(\mathbf{x}^{(i)}) \neq y^{(i)}]}}{n} \leq \prod_{m=1}^M \sqrt{1 - 4 (\hat{\gamma}^{[m]})^2}, \quad (1)$$

where $\hat{\gamma}^{[m]} = \frac{1}{2} - \text{err}^{[m]}$. For this purpose, proceed as follows:

(a) Give an interpretation of $\hat{\gamma}^{[m]}$.

Proof Roadmap: How the Five Parts Fit Together

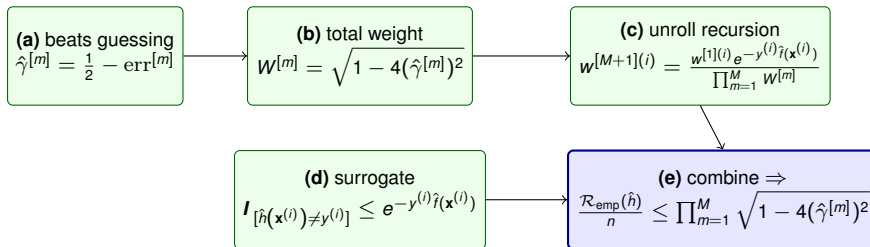
The bound is assembled in (e); parts (a)–(d) build the pieces it needs.



In (e): start from (d), rewrite via (c), bound each $W^{[m]}$ by (b).

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In (e): start from (d), rewrite via (c), bound each $W^{[m]}$ by (b).

Solution to Exercise 1 (a)

- ▶ Recall that $\text{err}^{[m]} = \sum_{i=1}^n w^{[m]}(i) \cdot I_{y^{(i)} \neq \hat{b}^{[m]}(\mathbf{x}^{(i)})}$ is the weighted error of $\hat{b}^{[m]}$.
- ▶ Random guessing (a fair coin flip) has weighted error exactly $\frac{1}{2}$.
- ▶ So, $\hat{\gamma}^{[m]} = \frac{1}{2} - \text{err}^{[m]}$ tells us how much *better* $\hat{b}^{[m]}$ is compared to random guessing.

Roadmap

▶ (a) $\hat{\gamma}^{[m]}$

better than guessing

(b) Weight $W^{[m]}$

$$= \sqrt{1 - 4(\hat{\gamma}^{[m]})^2}$$

(c) $w^{[M+1]}(i)$

unroll the recursion

(d) $I \leq e^{-y^t f}$

exp. surrogate

(e) Combine

(d) $\rightarrow$ (c) $\rightarrow$ (b)

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Exercise 1 (b)

(b) For any $m = 1, \dots, M$ let $W^{[m]} = \sum_{i=1}^n \tilde{w}^{[m](i)}$ be the total weight in iteration m before normalizing the weights. Show that $W^{[m]} = \sqrt{1 - 4(\hat{\gamma}^{[m]})^2}$.

Hint:

- ▶ $\tilde{w}^{[m](i)} = w^{[m](i)} \cdot \exp(-\beta^{[m]} y^{(i)} \hat{b}^{[m]}(\mathbf{x}^{(i)}))$.
- ▶ Two cases:
 - ▶ correct prediction: $y^{(i)} = \hat{b}^{[m]}(\mathbf{x}^{(i)}) \Leftrightarrow y^{(i)} \cdot \hat{b}^{[m]}(\mathbf{x}^{(i)}) = 1$
 - ▶ incorrect prediction: $y^{(i)} \neq \hat{b}^{[m]}(\mathbf{x}^{(i)}) \Leftrightarrow y^{(i)} \cdot \hat{b}^{[m]}(\mathbf{x}^{(i)}) = -1$
- ▶ $\text{err}^{[m]} = \sum_{i=1}^n w^{[m](i)} \cdot I_{y^{(i)} \neq \hat{b}^{[m]}(\mathbf{x}^{(i)})}$, i.e. $\text{err}^{[m]}$ sums the weights of all misclassified samples.

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1 (b): Think Before Solving

Target: $W^{[m]} = \sum_{i=1}^n \tilde{w}^{[m](i)} \stackrel{!}{=} \sqrt{1 - 4(\hat{\gamma}^{[m]})^2}.$

- **Split.** $y^{(i)} \hat{b}^{[m]}(\mathbf{x}^{(i)}) \in \{+1, -1\}$ only — can the sum be split into correct vs. incorrect samples?
- **Collapse.** Within each group, is the exponential factor the same for every sample? If so, what do $\sum_{\text{correct}} w^{[m](i)}$ and $\sum_{\text{incorrect}} w^{[m](i)}$ equal?
- **Substitute.** How does $\beta^{[m]} = \frac{1 - \text{err}^{[m]}}{2 \text{err}^{[m]}}$ turn $e^{\pm \beta^{[m]}}$ into a square root, and $\text{err}^{[m]}$ into $\hat{\gamma}^{[m]}$?

chain to build: $\hat{\gamma}^{[m]} \rightarrow \text{err}^{[m]} \rightarrow w^{[m](i)} \rightarrow \tilde{w}^{[m](i)} \rightarrow W^{[m]}$

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chain to build: $\hat{\gamma}^{[m]} \rightarrow \text{err}^{[m]} \rightarrow w^{[m](i)} \rightarrow \tilde{w}^{[m](i)} \rightarrow W^{[m]}$

1 (b): Split the Weight Sum into Two Groups

$y^{(i)} \hat{b}^{[m]}(\mathbf{x}^{(i)}) = +1$ on correct samples, $= -1$ on incorrect ones

$\Rightarrow$ split $\sum_{i=1}^n \tilde{w}^{[m](i)}$:

correct: $y^{(i)} = \hat{b}^{[m]}(\mathbf{x}^{(i)})$

$$\sum_{i: y^{(i)} = \hat{b}^{[m]}(\mathbf{x}^{(i)})} w^{[m](i)} = 1 - \text{err}^{[m]}$$

weight $\times e^{-\beta^{[m]}}$

$$\rightarrow e^{-\beta^{[m]}} (1 - \text{err}^{[m]})$$

incorrect: $y^{(i)} \neq \hat{b}^{[m]}(\mathbf{x}^{(i)})$

$$\sum_{i: y^{(i)} \neq \hat{b}^{[m]}(\mathbf{x}^{(i)})} w^{[m](i)} = \text{err}^{[m]}$$

weight $\times e^{+\beta^{[m]}}$

$$\rightarrow e^{+\beta^{[m]}} \text{err}^{[m]}$$

Roadmap

✓ (a) $\hat{\gamma}^{[m]}$

better than guessing

► (b) Weight $w^{[m]}$

$$= \sqrt{1 - 4(\hat{\gamma}^{[m]})^2}$$

(c) $w^{[M+1]}(i)$

unroll the recursion

(d) $I \leq e^{-y^f}$

exp. surrogate

(e) Combine

(d) $\rightarrow$ (c) $\rightarrow$ (b)

1 (b): Express $W^{[m]}$ with $w^{[m](i)}$, and then $\text{err}^{[m]}$

$$\begin{aligned} W^{[m]} &= \sum_{i=1}^n \tilde{w}^{[m](i)} \\ &= \sum_{i=1}^n w^{[m](i)} \exp \left(-\beta^{[m]} y^{(i)} \hat{b}^{[m]}(\mathbf{x}^{(i)}) \right) \end{aligned}$$

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1 (b): Substitute $\beta^{[m]}$ with An Expression of $\text{err}^{[m]}$

Summarizing the previous steps:

$$W^{[m]} = \sum_{i=1}^n \tilde{w}^{[m](i)} = \exp(\beta^{[m]}) \text{err}^{[m]} + \exp(-\beta^{[m]}) (1 - \text{err}^{[m]}) \quad (2)$$

Recall that $\beta^{[m]} = \frac{1}{2} \log \left(\frac{1 - \text{err}^{[m]}}{\text{err}^{[m]}} \right)$, so that

$$\exp(\beta^{[m]}) = \sqrt{\frac{1 - \text{err}^{[m]}}{\text{err}^{[m]}}}, \quad (3)$$

$$\exp(-\beta^{[m]}) = \sqrt{\frac{\text{err}^{[m]}}{1 - \text{err}^{[m]}}}. \quad (4)$$

We can then plug (3)–(4) into (2) and eliminate the terms related to $\beta^{[m]}$.

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(e) Combine

(d) $\rightarrow$ (c) $\rightarrow$ (b)

1 (b): Plug in the Relation between $\text{err}^{[m]}$ and $\hat{\gamma}^{[m]}$

$$\begin{aligned} W^{[m]} &= \exp(\beta^{[m]}) \text{err}^{[m]} + \exp(-\beta^{[m]}) (1 - \text{err}^{[m]}) \\ &= 2\sqrt{(1 - \text{err}^{[m]})\text{err}^{[m]}} \end{aligned}$$

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Exercise 1 (c)

(c) Show that

$$w^{[M+1]}(i) = \frac{w^{[1]}(i) \exp(-y^{(i)} \hat{f}(\mathbf{x}^{(i)}))}{\prod_{m=1}^M w^{[m]}},$$

where $w^{[M+1]}(i)$ is the **normalized** weight we would obtain if AdaBoost were run for $M + 1$ iterations.

Hint:

$$w^{[m+1]}(i) = \frac{\tilde{w}^{[m]}(i)}{\sum_{i=1}^n \tilde{w}^{[m]}(i)} = \frac{w^{[m]}(i) \cdot \exp(-\beta^{[m]} y^{(i)} \hat{b}^{[m]}(\mathbf{x}^{(i)}))}{\sum_{i=1}^n w^{[m]}(i) \cdot \exp(-\beta^{[m]} y^{(i)} \hat{b}^{[m]}(\mathbf{x}^{(i)}))}$$

- The above hint relates $w^{[m+1]}(i)$ to $w^{[m]}(i)$, or $w^{[m]}(i)$ to $w^{[m-1]}(i)$, ..., or $w^{[2]}(i)$ to $w^{[1]}(i)$. This motivates a recursive proof.

1 (c): Think Before Solving

Target: $w^{[M+1]}(i) = \frac{w^{[1]}(i) \exp(-y^{(i)} \hat{f}(\mathbf{x}^{(i)}))}{\prod_{m=1}^M W^{[m]}}.$

- ▶ **One step.** The hint writes $w^{[m+1]}(i)$ via $w^{[m]}(i)$ and $W^{[m]}$ — can you unfold this from level $M+1$ down to level 1?
- ▶ **Stacking.** Once the steps are stacked, what happens to the $W^{[m]}$ factors and to the product of exponentials?
- ▶ **Exponent sum.** Why does $\sum_m \beta^{[m]} \hat{b}^{[m]}(\mathbf{x}^{(i)})$ collapse into the single score $\hat{f}(\mathbf{x}^{(i)})$?

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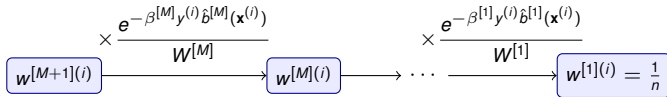
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1 (c): Unrolling the Weight Recursion

Apply $w^{[m+1]}(i) = w^{[m]}(i) \frac{e^{-\beta^{[m]} y^{(i)} \hat{b}^{[m]}(\mathbf{x}^{(i)})}}{W^{[m]}}$ repeatedly, peeling one factor each step:



All M exponentials multiply; all M weights $W^{[m]}$ collect in the denominator.

Roadmap

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✓ (b) Weight $W^{[m]}$
 $= \sqrt{1 - 4(\hat{\gamma}^{[m]})^2}$

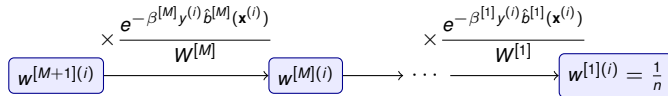
► (c) $w^{[M+1]}(i)$
unroll the recursion

(d) $I \leq e^{-y^t}$
exp. surrogate

(e) Combine
(d) $\rightarrow$ (c) $\rightarrow$ (b)

1 (c): Unrolling the Weight Recursion

Apply $w^{[m+1]}(i) = w^{[m]}(i) \frac{e^{-\beta^{[m]} y^{(i)} \hat{b}^{[m]}(\mathbf{x}^{(i)})}}{W^{[m]}}$ repeatedly, peeling one factor each step:



All M exponentials multiply; all M weights $W^{[m]}$ collect in the denominator.

Roadmap

✓ (a) $\hat{\gamma}^{[m]}$
better than guessing

✓ (b) Weight $W^{[m]}$
 $= \sqrt{1 - 4(\hat{\gamma}^{[m]})^2}$

► (c) $w^{[M+1]}(i)$
unroll the recursion

(d) $I \leq e^{-y^f}$
exp. surrogate

(e) Combine
(d) $\rightarrow$ (c) $\rightarrow$ (b)

1 (c): From $w^{[M+1]}(i)$ to $w^{[M]}(i)$, and to $w^{[M-1]}(i)$

Note: the target involves $\prod_{m=1}^M W^{[m]}$ in the denominator.

$$\begin{aligned}w^{[M+1]}(i) &= w^{[M]}(i) \cdot \frac{\exp(-\beta^{[M]} y^{(i)} \hat{b}^{[M]}(\mathbf{x}^{(i)}))}{\sum_{i=1}^n w^{[M]}(i) \cdot \exp(-\beta^{[M]} y^{(i)} \hat{b}^{[M]}(\mathbf{x}^{(i)}))} \\&= w^{[M]}(i) \cdot \frac{\exp(-\beta^{[M]} y^{(i)} \hat{b}^{[M]}(\mathbf{x}^{(i)}))}{W^{[M]}} \quad (\text{Def. of } W^{[M]})\end{aligned}$$

Roadmap

✓ (a) $\hat{\gamma}^{[m]}$

better than guessing

✓ (b) Weight $W^{[m]}$

$$= \sqrt{1 - 4(\hat{\gamma}^{[m]})^2}$$

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Roadmap

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(e) Combine

(d) $\rightarrow$ (c) $\rightarrow$ (b)

Exercise 1 (d)

(d) Argue that $I_{[\hat{h}(\mathbf{x}) \neq y]} \leq \exp(-y\hat{f}(\mathbf{x}))$ for any $(\mathbf{x}, y) \in \mathcal{X} \times \mathcal{Y}$.

Hint: when $\hat{h}(\mathbf{x}) \neq y$ (the indicator = 1), how large is $\exp(-y\hat{f}(\mathbf{x}))$?

1 (d): Think Before Solving

Target: $I_{[\hat{h}(\mathbf{x}) \neq y]} \leq \exp(-y\hat{f}(\mathbf{x}))$ for all $(\mathbf{x}, y)$.

- ▶ **Two outcomes.** The indicator is 0 or 1 — check the inequality separately for a correct and an incorrect prediction.
- ▶ **Misclassified.** $\hat{h} = \text{sign}(\hat{f})$, so $\hat{h}(\mathbf{x}) \neq y \Leftrightarrow y\hat{f}(\mathbf{x}) < 0$. What does that force $\exp(-y\hat{f}(\mathbf{x}))$ to exceed?
- ▶ **Picture.** Sketch $e^{-\nu}$ and the 0/1 step against $\nu = yf$ — does the curve ever dip below the step?

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- ▶ **Picture.** Sketch $e^{-\nu}$ and the 0/1 step against $\nu = yf$ — does the curve ever dip below the step?

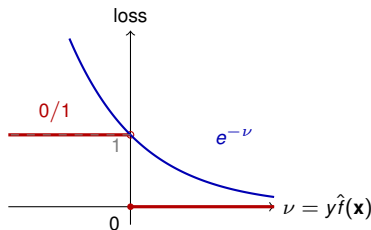
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1 (d): The Exponential Upper-Bounds the 0/1 Loss

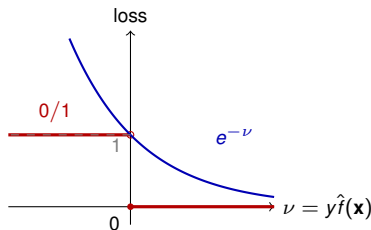
- ▶ Write the margin $\nu := y\hat{f}(\mathbf{x})$.
- ▶ 0/1 loss $I_{[\hat{h} \neq y]} = I_{[\nu < 0]}$ (step at 0).
- ▶ Surrogate $e^{-\nu}$ lies *above* the step for *all* ν :
 - ▶ $\nu < 0$ (wrong): $e^{-\nu} > 1 = I$.
 - ▶ $\nu \geq 0$ (right): $e^{-\nu} > 0 = I$.
- ▶ $\Rightarrow I_{[\hat{h} \neq y]} \leq \exp(-y\hat{f}(\mathbf{x}))$.



$$\begin{aligned}\hat{h}(\mathbf{x}) \neq y &\Leftrightarrow \text{sign}(\hat{f}(\mathbf{x})) \neq y \Leftrightarrow -y\hat{f}(\mathbf{x}) > 0 \\ &\Leftrightarrow \exp(-y\hat{f}(\mathbf{x})) > \exp(0) = 1 = I_{[\hat{h}(\mathbf{x}) \neq y]}.\end{aligned}$$

1 (d): The Exponential Upper-Bounds the 0/1 Loss

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Exercise 1 (e)

(e) Combine everything to conclude

$$\frac{\mathcal{R}_{\text{emp}}(\hat{h})}{n} = \frac{\sum_{i=1}^n \mathbb{I}_{[\hat{h}(\mathbf{x}^{(i)}) \neq y^{(i)}]}}{n} \leq \prod_{m=1}^M \sqrt{1 - 4 (\hat{\gamma}^{[m]})^2}.$$

Hint:

- ▶ In (b), we proved $W^{[m]} = \sqrt{1 - 4 (\hat{\gamma}^{[m]})^2}$.
- ▶ In (c), $w^{[M+1]}(i) = \frac{w^{[1]}(i) \exp(-y^{(i)} \hat{f}(\mathbf{x}^{(i)}))}{\prod_{m=1}^M W^{[m]}}$.
- ▶ In (d): $\mathbb{I}_{[\hat{h}(\mathbf{x}) \neq y]} \leq \exp(-y \hat{f}(\mathbf{x}))$

1 (e): Think Before Solving

Target: $\frac{\mathcal{R}_{\text{emp}}(\hat{h})}{n} \leq \prod_{m=1}^M \sqrt{1 - 4(\hat{\gamma}^{[m]})^2}.$

- **Start.** The error is an average of indicators — which earlier bound replaces each indicator?
- **Reweight.** With $w^{[1]}(i) = \frac{1}{n}$, which identity rewrites $w^{[1]}(i) e^{-y^{(i)} \hat{f}(\mathbf{x}^{(i)})}$ as $w^{[M+1]}(i) \prod_{m=1}^M W^{[m]}$?
- **Close.** Why does $\sum_{i=1}^n w^{[M+1]}(i) = 1$, and which result turns $\prod_{m=1}^M W^{[m]}$ into the final square-root product?

$$(d) I \leq e^{-y^f}$$

$$(c) w^{[1]}(i) e^{-y^f} = w^{[M+1]}(i) \prod_{m=1}^M W^{[m]}$$

$$(b) W^{[m]} = \sqrt{1 - 4(\hat{\gamma}^{[m]})^2}$$

1 (e): Think Before Solving

Target: $\frac{\mathcal{R}_{\text{emp}}(\hat{h})}{n} \leq \prod_{m=1}^M \sqrt{1 - 4(\hat{\gamma}^{[m]})^2}.$

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$$(d) I \leq e^{-yf}$$

$$(c) w^{[1](i)} e^{-yf} = w^{[M+1](i)} \prod_{m=1}^M W^{[m]}$$

$$(b) W^{[m]} = \sqrt{1 - 4(\hat{\gamma}^{[m]})^2}$$

1 (e): Use (d), and then (c) and then (b)

$$\begin{aligned}\frac{\mathcal{R}_{\text{emp}}(\hat{h})}{n} &= \frac{\sum_{i=1}^n I_{\hat{h}(\mathbf{x}^{(i)}) \neq y^{(i)}}}{n} = \sum_{i=1}^n \frac{1}{n} \cdot I_{\hat{h}(\mathbf{x}^{(i)}) \neq y^{(i)}} \\ &\leq \sum_{i=1}^n \frac{1}{n} \cdot \exp\left(-y^{(i)} \hat{f}(\mathbf{x}^{(i)})\right) \quad \text{(d)}\end{aligned}$$

1 (e): Use (d), and then (c) and then (b)

$$\begin{aligned}\frac{\mathcal{R}_{\text{emp}}(\hat{h})}{n} &= \frac{\sum_{i=1}^n I_{\hat{h}(\mathbf{x}^{(i)}) \neq y^{(i)}}}{n} = \sum_{i=1}^n \frac{1}{n} \cdot I_{\hat{h}(\mathbf{x}^{(i)}) \neq y^{(i)}} \\ &\leq \sum_{i=1}^n \frac{1}{n} \cdot \exp\left(-y^{(i)} \hat{f}(\mathbf{x}^{(i)})\right) \quad \text{(d)} \\ &= \sum_{i=1}^n w^{[1](i)} \exp\left(-y^{(i)} \hat{f}(\mathbf{x}^{(i)})\right) \quad (\text{Definition of } w^{[1](i)} = 1/n)\end{aligned}$$

1 (e): Use (d), and then (c) and then (b)

$$\begin{aligned}\frac{\mathcal{R}_{\text{emp}}(\hat{h})}{n} &= \frac{\sum_{i=1}^n \mathbf{1}_{\hat{h}(\mathbf{x}^{(i)}) \neq y^{(i)}}}{n} = \sum_{i=1}^n \frac{1}{n} \cdot \mathbf{1}_{\hat{h}(\mathbf{x}^{(i)}) \neq y^{(i)}} \\ &\leq \sum_{i=1}^n \frac{1}{n} \cdot \exp\left(-y^{(i)} \hat{f}(\mathbf{x}^{(i)})\right) \quad \text{(d)} \\ &= \sum_{i=1}^n w^{[1](i)} \exp\left(-y^{(i)} \hat{f}(\mathbf{x}^{(i)})\right) \quad \text{(Definition of } w^{[1](i)} = 1/n\text{)} \\ &= \sum_{i=1}^n w^{[M+1](i)} \prod_{m=1}^M w^{[m]} \quad \text{(c): } w^{[M+1](i)} = \frac{w^{[1](i)} \exp(-y^{(i)} \hat{f}(\mathbf{x}^{(i)}))}{\prod_{m=1}^M w^{[m]}}\end{aligned}$$

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The training error decays geometrically as long as every round beats guessing ($\hat{\gamma}^{[m]} > 0$).



Summary — Connecting the exercise to the lecture

What we derived	Lecture concept made concrete
(a) $\hat{\gamma}^{[m]} = \frac{1}{2} - \text{err}^{[m]}$ measures how far the weighted error of $\hat{b}^{[m]}$ beats random guessing.	Weighted base-learner error; the AdaBoost weighting scheme.
(b) Total unnormalized weight $W^{[m]} = \sqrt{1 - 4(\hat{\gamma}^{[m]})^2}$ (via $\beta^{[m]} = \frac{1}{2} \log \frac{1 - \text{err}^{[m]}}{\text{err}^{[m]}}$).	AdaBoost weight update $\beta^{[m]}$; normalizing constant per round.
(c) Unrolling the reweighting recursion gives $w^{[M+1](i)} = w^{[1](i)} e^{-y^{(i)} \hat{f}(\mathbf{x}^{(i)})} / \prod_{m=1}^M W^{[m]}$, linking single weights to the ensemble score $\hat{f}$.	Additive model $\hat{f} = \sum_m \beta^{[m]} \hat{b}^{[m]}$; multiplicative weight recursion.
(d) $I_{[\hat{h} \neq y]} \leq \exp(-y \hat{f}(\mathbf{x}))$: the exponential is a convex <i>surrogate</i> upper-bounding the 0/1 loss.	Exponential loss as a surrogate for misclassification loss.
(e) Combine (d) $\rightarrow$ (c) $\rightarrow$ (b): $\frac{\mathcal{R}_{\text{emp}}(\hat{h})}{n} \leq \prod_{m=1}^M \sqrt{1 - 4(\hat{\gamma}^{[m]})^2}$.	Training-error bound; boosting drives error down if every $\hat{\gamma}^{[m]} > 0$.